Prior-Art Pass Report for `p-vs-np`

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Abstract

This report records a targeted prior-art pass over the current blocked proof frontier. It imports decisive external theorem-level references into the problem-scoped MathLib and explains what those references change about the current proof state.

1 Snapshot

- Imported references: 8
- Blocked queue items: 24
- Covered blocked queue items: 24
- Uncovered blocked queue items:

2 Search Justification

The local queue is exhausted through route conjectures, abstraction repairs, and repair lemmas, so browsing is justified only for named missing lemmas that materially affect novelty or barrier judgments.

3 Key Findings

- The imported references ground most target-side theorems and barriers, but none closes the missing carrier-realization or formula-size transport lemmas.
- The PIT route sharpens from a vague semantic-to-algorithmic gap into an explicitness gap: known results already connect explicit hitting sets to black-box PIT and PIT derandomization to lower bounds.
- The Cook-Reckhow route is not discovering a new target theorem; the real gap is the bridge from the local definitional object to the existing NP=coNP characterization.
- The MCSP local-reduction route is now barrier-aware because prior art already shows non-NP-hardness under local reductions and strong consequences for harder reductions.
- The general magnification route is literature-backed, but the unresolved step remains object-class alignment between the MCSP-specific carrier and the sparse-problem or distinguisher framework.

4 Imported References

ID	Type	Title	Obstruction classes
mathlibref.p-vs-np.001	theorem	Hardness Magnification for Natural Problems	carrier-realization, mcsp-magnification
mathlibref.p-vs-np.002	theorem	Hardness Magnification Near State-of-the-Art Lower Bounds	carrier-realization, formula-size-transport, mcsp-magnification
mathlibref.p-vs-np.003	theorem	MCSP Is Not NP-Hard Under Local Reductions	local-reduction-barrier
mathlibref.p-vs-np.004	theorem	Hitting-Set Generators Yield Black-Box PIT	pit-upgrade, explicitness-gap
mathlibref.p-vs-np.005	theorem	Derandomizing Polynomial Identity Tests Means Proving Circuit Lower Bounds	pit-upgrade, target-grounding
mathlibref.p-vs-np.008	theorem	Many Natural Problems Are NP-Complete	schema-to-examples, target-grounding
mathlibref.p-vs-np.006	theorem	Polynomially Bounded Proof Systems Characterize NP Equals coNP	characterization-bridge, target-grounding
mathlibref.p-vs-np.007	theorem	A Simple and General Hardness Magnification Theorem	target-alignment, general-magnification

5 Obstruction Map

carrier-realization

References. mathlibref.p-vs-np.001, mathlibref.p-vs-np.002

Queue ids. proofq.p-vs-np.001, proofq.p-vs-np.002, proofq.p-vs-np.005, proofq.p-vs-np.009, proofq.p-vs-np.010, proofq.p-vs-np.013, proofq.p-vs-np.018, proofq.p-vs-np.020, proofq.p-vs-np.021

- The leading MCSP route is no longer novel at the level of magnification-as-such; the open step is the typed carrier or realization theorem that would instantiate the route inside the current proof state.
- The bridge.001 target is externally grounded, so the remaining novelty claim shifts to how the current framework realizes a usable MCSP carrier rather than to the target theorem itself.

characterization-bridge

References. mathlibref.p-vs-np.006

Queue ids. proofq.p-vs-np.006, proofq.p-vs-np.014, proofq.p-vs-np.024

- The Cook-Reckhow route is now clearly a bridge problem, not a target-theorem problem: the open step is the semantic bridge from the local definitional object to this existing characterization theorem.

explicitness-gap

References. mathlibref.p-vs-np.004
Queue ids. proofq.p-vs-np.016, proofq.p-vs-np.017

- The PIT route is not blocked by the total absence of a hitting-set-to-PIT connection; it is blocked by the lack of an explicit generator or certificate object for the right class.

formula-size-transport

References. mathlibref.p-vs-np.002
Queue ids. proofq.p-vs-np.001, proofq.p-vs-np.005, proofq.p-vs-np.009, proofq.p-vs-np.010, proofq.p-vs-np.013, proofq.p-vs-np.018, proofq.p-vs-np.020, proofq.p-vs-np.021

- The bridge.001 target is externally grounded, so the remaining novelty claim shifts to how the current framework realizes a usable MCSP carrier rather than to the target theorem itself.

general-magnification

References. mathlibref.p-vs-np.007
Queue ids. proofq.p-vs-np.003, proofq.p-vs-np.011, proofq.p-vs-np.023

- The general magnification route is literature-backed, but its live gap is an alignment theorem from the MCSP obstruction family to the sparse-problem or distinguisher object class required by the general theorem.

local-reduction-barrier

References. mathlibref.p-vs-np.003
Queue ids. proofq.p-vs-np.004, proofq.p-vs-np.012, proofq.p-vs-np.019

- The local-reduction route is now explicitly barrier-aware: the missing gadget lemma must overcome known non-hardness and consequence results, not just fill an arbitrary construction gap.

mcsp-magnification

References. mathlibref.p-vs-np.001, mathlibref.p-vs-np.002
Queue ids. proofq.p-vs-np.001, proofq.p-vs-np.002, proofq.p-vs-np.005, proofq.p-vs-np.009, proofq.p-vs-np.010, proofq.p-vs-np.013, proofq.p-vs-np.018, proofq.p-vs-np.020, proofq.p-vs-np.021

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pit-upgrade

References. mathlibref.p-vs-np.004, mathlibref.p-vs-np.005
Queue ids. proofq.p-vs-np.008, proofq.p-vs-np.016, proofq.p-vs-np.017

- The PIT route is not blocked by the total absence of a hitting-set-to-PIT connection; it is blocked by the lack of an explicit generator or certificate object for the right class.
- The PIT route's target theorem is prior art; the remaining open step is the current framework's inability to produce the required derandomization premise from its local PIT carrier.

schema-to-examples

References. mathlibref.p-vs-np.008
Queue ids. proofq.p-vs-np.007, proofq.p-vs-np.015, proofq.p-vs-np.022

- The NP-complete examples route is now target-grounded: the open step is constructing or transferring explicit examples from the local abstraction, not discovering that natural NP-complete problems exist.

target-alignment

References. mathlibref.p-vs-np.007
Queue ids. proofq.p-vs-np.003, proofq.p-vs-np.011, proofq.p-vs-np.023

- The general magnification route is literature-backed, but its live gap is an alignment theorem from the MCSP obstruction family to the sparse-problem or distinguisher object class required by the general theorem.

target-grounding

References. mathlibref.p-vs-np.005, mathlibref.p-vs-np.006, mathlibref.p-vs-np.008
Queue ids. proofq.p-vs-np.006, proofq.p-vs-np.007, proofq.p-vs-np.008, proofq.p-vs-np.014, proofq.p-vs-np.015, proofq.p-vs-np.016, proofq.p-vs-np.017, proofq.p-vs-np.022, proofq.p-vs-np.024

- The PIT route's target theorem is prior art; the remaining open step is the current framework's inability to produce the required derandomization premise from its local PIT carrier.
- The NP-complete examples route is now target-grounded: the open step is constructing or transferring explicit examples from the local abstraction, not discovering that natural NP-complete problems exist.
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6 Still Uncovered

- All blocked queue items have at least one linked imported reference.